NANDHA GOPAL M

Embedded Firmware Engineer | ESP32 · STM32 · FreeRTOS · WiFi/TCP

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PROFESSIONAL SUMMARY

Embedded firmware engineer (fresher) who has built industrial controller firmware on ESP32 and STM32 (FreeRTOS, custom serial/WiFi protocols, LVGL, WiFi/TCP). Strong in board bring-up, hardware-software integration and bench debugging, backed by two years of board-level fault diagnosis as a service technician. Available immediately.

TECHNICAL SKILLS

Languages: Embedded C · C++ (basic) · Python · Bash

RTOS & Real-Time: FreeRTOS (tasks, queues, semaphores) · Hardware timers · Interrupts · Finite state machines

Peripherals & Protocols: GPIO · ADC · PWM · UART · SPI · I2C · WiFi (AP mode, mDNS) · TCP server

Hardware & Debugging: Board bring-up · HW-SW integration · Serial / black-box logging · KiCad (PCB)

Hardware Tools: SMD rework station · Soldering (SMD / through-hole) · DC bench power supply · Multimeter · Oscilloscope · Logic analyzer · Microscope · Thermal camera (short finding)

Platforms & Tools: STM32 · ESP32 / ESP32-S3 · Arduino Mega · Raspberry Pi · LVGL · PlatformIO · Git

EXPERIENCE

Navson Technologies — Embedded Firmware Developer (Intern) Jun 2026 – Aug 2026

- Delivered embedded firmware and hardware for ESP32-based industrial process-control systems (dip coating, spray coating): motor control, sensor integration and communication over UART, SPI, I2C, GPIO and WiFi.
- Worked end-to-end across backend controller firmware, HMI development and PCB/prototype hardware design, taking systems from prototype through production and on-site field support.

Project 1 — Dip Coater Backend Controller — ESP32, FreeRTOS, Custom WiFi/Serial Protocol, KiCad

- Designed and built the backend controller for an industrial dip coater: KiCad schematic and a prototype board (breadboard/dot-board model) driving a TB6600 + NEMA23 stepper stage with ADC sensor feedback and defined motion boundaries.
- During machine testing, replaced the polling-based FreeRTOS backend logic with an interrupt/event-driven design (hardware timer + ISR-set atomic flags) for more reliable real-time control.
- Designed a custom CSV command protocol shared identically over USB Serial and WiFi TCP, with mDNS-based automatic device discovery and a UDP broadcast fallback.
- Diagnosed and fixed hardware issues found on the machine; the controller firmware and board were moved into production on the dip coater line.

Project 2 — Dip Coater HMI & Wireless Upgrade — ESP32-S3, LVGL, WiFi

- Designed and built a 7-inch touchscreen HMI on ESP32-S3 using LVGL to meet a lower-cost, industrial-grade product requirement, replacing the client's earlier Bluetooth + Arduino Mega interface.
- Implemented new WiFi-based communication between the HMI (ESP32-S3) and backend controller (ESP32), replacing the Mega link with a dual-ESP32 industrial-grade architecture.
- Integrated the new HMI end-to-end with the dip coater application.

Project 3 — Field Support: Spray Coater XY Gantry Controller — Arduino Mega, TB6600, Nextion HMI

- Provided on-site fault support for a spray coater sold two years earlier and heavily used in the field, with no schematic available on site.
- Root-caused unstable X/Y motion by validating every sensor line at the JST-XH connector with a multimeter and a known-good external sensor: found miswired signal pins, a voltage mix-up that had burnt a GND line, and a missing common ground; restored the wiring on a replacement board.
- Fixed the firmware side: corrected the endstop state check for the reverse-direction condition and a parameter-passing bug, plus resolved NC vs. NO wiring and INPUT_PULLUP configuration mismatches.

Xiaomi Authorised Service Centre — Service Technician 2021 – 2023

- Diagnosed and repaired motherboard-level failures in Android smartphones (power ICs, charging circuits) on dense multi-layer boards using measurement and test-point probing.
- Performed firmware flashing, software recovery and data restoration for bricked devices; this systematic fault isolation is the approach I now apply to embedded debugging.

PERSONAL & ACADEMIC PROJECTS

Driver Safety Monitoring System (Final Year Project) — STM32, Embedded C, OpenCV, MediaPipe, Python, UART

- Built a driver-safety system combining vision-based drowsiness detection (OpenCV + MediaPipe: Eye Aspect Ratio and PERCLOS) with STM32-side alcohol sensing (MQ-3 → ADC → threshold logic), linked over UART.
- Implemented automated response control (buzzer alert, motor cut-off) on drowsiness/alcohol detection, plus a Tkinter GUI for real-time monitoring and data logging.
- Validated threshold-based detection and serial data integrity end-to-end, and documented the faults found and their fixes.

Real-Time Phishing Detection & Blocking System — Raspberry Pi, Pi-hole, Bash, Python

Raspberry Pi running Pi-hole as a network-level DNS filter, with Bash scripts for automated blocklist updates and health monitoring.

EDUCATION

B.E. – Electronics and Communication Engineering, PSR Engineering College (CGPA: 7.5/10) 2023 – 2026

CERTIFICATIONS & DOCUMENTS

NPTEL – Embedded System Design (2025) · Data Structures and Algorithms – FinestCoder (2024)

Internship Certificate (click to view) | Letter of Recommendation (click to view)